

Exact Polynomial Plateaux from Rvachev’s Up-Function

Dyadic oversampling, Appell deconvolution, finite interval algorithms,
Thue–Morse antiderivatives, and sharp reproduction order

Research report prepared for Vladimir Reshetnikov

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Abstract

Let up denote Rvachev’s even C^∞ compactly supported function on $[-1, 1]$, normalized by $\int \text{up} = 1$. This report gives an explicit exact solution of the following problem: for an arbitrary polynomial p and a prescribed compact interval $I = [A, B]$, construct a finite sum of shifted and scaled copies of up agreeing with p at every point of I .

The construction is stronger than existence. All atoms may have one common horizontal scale. Choose an integer oversampling ratio m with $\nu_2(m) \geq \deg p$, a lattice step $h > 0$, and centers $x_k = x_0 + kh$. Put $\rho = mh$, let

$$M(z) = \int_{\mathbb{R}} e^{zx} \text{up}(x) \, dx = \prod_{j=1}^{\infty} \frac{\sinh(z/2^j)}{z/2^j},$$

and define the polynomial deconvolution

$$p_{\rho}^{\#} = M(\rho D)^{-1} p.$$

Then the locally finite identity

$$p(x) = \frac{1}{m} \sum_{k \in \mathbb{Z}} p_{\rho}^{\#}(x_0 + kh) \text{up}\left(\frac{x - x_0 - kh}{\rho}\right)$$

holds on all of $\mathbb{R}$. Keeping only atoms whose supports meet I gives the requested finite formula. The inverse operator terminates on polynomials and has rational coefficients expressible through Bernoulli cumulants and complete Bell polynomials; no collocation system or numerical inversion is needed.

A twisted Poisson argument proves that the exact stationary reproduction degree is precisely $\nu_2(m)$. Hence the smallest integer ratio in this common-scale lattice scheme is $m = 2^{\deg p}$. The first failed degree has an explicit nonzero Fourier-series defect. A complementary real-space construction repeatedly integrates finite up-sums through the scale-difference identity for up' ; invisible “ghost” atoms repair the finite zero-sum compatibility conditions and control constants of integration. Iteration yields an exact Thue–Morse derivative stencil. The report also treats finite sinc products, arithmetic, support, conditioning, multivariate tensor products, conjectures, and future directions. The accompanying Python program uses exact rational arithmetic: its 49-atom degree-four example has residual exactly zero at 129 dyadic points, and seven additional degree tests from zero through six also return exact zero.

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1 Scope, corpus audit, and status of the results

1.1 Repository corpus examined

The final pinned repository snapshot contains 97 LaTeX sources under `Analysis/FabiusFunction/docs`, at commit

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The tree advanced during preparation, so the audit was refreshed from the earlier 85-source checkpoint. The twelve added sources comprise three independent dyadic-comb reports, two global-interpolation reports, and small generated or reconstructed TeX fragments. The inventory otherwise includes polished syntheses, archived versions, source-paper transcriptions, a mathematical glossary, Lean walkthroughs, and the large non-formalized frontier compilation. Corpus-wide searches and close reading covered:

- the normalizations of the bounded Fabius function, its signed global extension, and Rvachev’s up;
- support, positivity, endpoint flatness, and the functional-differential equation for up;
- the infinite sinc product for $\widehat{\text{up}}$, its integer zeros, and their exact dyadic multiplicities;
- Poisson summation, partition-of-unity identities, oversampled translate sums, shifted dyadic-comb quadrature, and alias defects;
- Kabaya–Iri and centered Rvachev–Appell polynomials, inverse moment operators, Bell–Bernoulli formulae, and self-sampling identities;
- exact dyadic values, Thue–Morse cancellation, finite convolution splines, interpolation, approximation, and frontier proposals.

The refreshed nonduplication boundary is important. The consolidated *Inverse and Sampling Frontiers* volume already develops the centered Rvachev–Appell inverse, links it to dyadic self-sampling, proves a shifted Strang–Fix exactness theorem, and analyzes the first alias obstruction [3]. Those results, together with the Fourier zero multiplicities, are treated here as corpus inputs rather than rediscoveries. The newly added comb papers concern quadrature of polynomials against up or Fabius weights, while the two interpolation papers concern global equispaced polynomial interpolants; neither constructs a finite sum of up-atoms that equals a prescribed arbitrary polynomial throughout a chosen interval.

Relative to that stronger prior-art boundary, the additions developed here are the interval-synthesis layer: the physical-scale formula for arbitrary p , explicit finite truncation indices, support and atom-count bounds, compressed coefficient generation, the sharp common-scale feasibility and complexity statements, the real-space ghost-antiderivative algorithm, its Thue–Morse derivative stencil, the finite-sinc-prefix transfer theorem, and tensor-product extension. The stationary Appell and alias formulae are re-derived in the report because they make the construction self-contained, but are labeled conceptually as prerequisites rather than corpus-relative novelty.

1.2 Priority and proof labels

Polynomial reproduction by integer translates is classical. The zero conditions used below belong to the Strang–Fix tradition [7, 8, 9, 10]. Novelty claims are therefore deliberately local: “not found in the inspected ProveIt corpus,” with emphasis on specialization to the unusual dyadic zero multiplicities of $\widehat{\text{up}}$, the exact coefficient formulas, and the interval algorithm. No exhaustive claim over the world literature is made.

Every numbered theorem, proposition, lemma, and corollary below has a proof or an explicit reduction to a standard cited theorem. Statements labeled *Conjecture* are unproved. Numerical computations only check identities already derived symbolically.

2 Normalization and repository inputs

Throughout,

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i x \xi} dx, \quad \text{sinc } z = \begin{cases} \sin z / z, & z \neq 0, \\ 1, & z = 0. \end{cases}$$

We write $\phi = \text{up}$ when emphasizing its role as a generator.

Proposition 2.1 (Basic data for up). *The repository normalization satisfies*

$$\phi \in C^\infty(\mathbb{R}), \quad \text{supp } \phi = [-1, 1], \quad \phi(-x) = \phi(x), \quad \int_{\mathbb{R}} \phi(x) dx = 1, \quad (1)$$

$$\phi'(x) = 2(\phi(2x+1) - \phi(2x-1)), \quad (2)$$

$$\widehat{\phi}(\xi) = \prod_{n=0}^{\infty} \text{sinc}\left(\frac{\pi \xi}{2^n}\right), \quad (3)$$

$$\sum_{k \in \mathbb{Z}} \phi(t+k) = 1. \quad (4)$$

All derivatives vanish at ± 1 , and $\phi > 0$ on $(-1, 1)$.

These facts are developed in the repository synthesis and by Arias de Reyna [5]. Every translate sum in this report is locally finite because of compact support.

Proposition 2.2 (Integer zero multiplicities). *For every nonzero integer q , $\widehat{\phi}$ has a zero of exact order*

$$1 + \nu_2(q)$$

at q .

Proof. The factor indexed by n in (3) vanishes at q exactly when $2^n \mid q$. Thus the vanishing factors are $n = 0, 1, \dots, \nu_2(q)$; each zero is simple and all remaining factors are nonzero. $\square$

Corollary 2.3 (Oversampled partition). *For every positive integer m ,*

$$\sum_{k \in \mathbb{Z}} \phi\left(\frac{t-k}{m}\right) = m.$$

Proof. Poisson summation gives Fourier modes $m\widehat{\phi}(mn)e^{-2\pi i n t}$. The zero mode is m , and all nonzero modes vanish by Theorem 2.2. $\square$

3 The exact interval problem

3.1 What is and is not possible

Given $p \in \mathbb{R}[x]$ and $I = [A, B]$, seek constants c_j , centers b_j , and scales $a_j > 0$ such that

$$p(x) = \sum_{j=1}^N c_j \phi\left(\frac{x-b_j}{a_j}\right) \quad (x \in I). \quad (5)$$

Equality means functional equality at every point, not interpolation at a finite set of nodes.

Proposition 3.1 (Global finite-support obstruction). *A finite sum of shifted and scaled copies of ϕ cannot equal a nonzero polynomial on all of $\mathbb{R}$.*

Proof. A finite sum of compactly supported functions is compactly supported, whereas a nonzero polynomial is not. $\square$

The interval is therefore essential. The correct global object is an infinite but locally finite lattice sum; compact support makes it finite after restriction to a compact interval.

3.2 A stronger target

All scales will be equal, all centers will lie on one arithmetic lattice, and all coefficients will be samples of one explicitly computable polynomial. Put

$$x_k = x_0 + kh, \quad \rho = mh,$$

where $h > 0$ is the center spacing and $m \in \mathbb{N}_{>0}$ is the integer ratio of atom radius to spacing. The atoms are

$$\phi_{\rho,k}(x) = \phi\left(\frac{x - x_k}{\rho}\right).$$

At any fixed x , at most $2m + 1$ atoms are nonzero.

4 Moment-generating product and Bernoulli cumulants

Define

$$M(z) = \int_{\mathbb{R}} e^{zx} \phi(x) \, dx. \quad (6)$$

Substitution $\xi = iz/(2\pi)$ into (3) gives

$$M(z) = \prod_{j=1}^{\infty} \frac{\sinh(z/2^j)}{z/2^j}. \quad (7)$$

The function is even, entire, and satisfies $M(0) = 1$.

Proposition 4.1 (Cumulants). *If*

$$\log M(z) = \sum_{r=1}^{\infty} \kappa_{2r} \frac{z^{2r}}{(2r)!},$$

then

$$\boxed{\kappa_{2r} = \frac{2^{2r-1} B_{2r}}{r(2^{2r} - 1)}} \quad (r \geq 1). \quad (8)$$

In particular,

$$\kappa_2 = \frac{1}{9}, \quad \kappa_4 = -\frac{2}{225}, \quad \kappa_6 = \frac{16}{3969}, \quad \kappa_8 = -\frac{16}{3825}.$$

Proof. Use

$$\log \frac{\sinh w}{w} = \sum_{r \geq 1} \frac{2^{2r-1} B_{2r}}{r(2r)!} w^{2r},$$

insert $w = z/2^j$, and sum $\sum_{j \geq 1} 2^{-2rj} = 1/(2^{2r} - 1)$. $\square$

Remark 4.2 (Odd denominators). After reduction, every κ_{2r} has odd denominator. The reduced denominator of B_{2r} contains exactly one factor of two, and $2^{2r-1}/r$ cancels it. This yields a clean 2-adic bound for the reproduction coefficients.

Let $\mu_j = \int x^j \phi(x) \, dx$. Moments follow from cumulants by

$$\mu_0 = 1, \quad \mu_n = \sum_{j=1}^n \binom{n-1}{j-1} \kappa_j \mu_{n-j}, \quad (9)$$

with odd κ_j and μ_j zero. The first values are

$$\mu_2 = \frac{1}{9}, \quad \mu_4 = \frac{19}{675}, \quad \mu_6 = \frac{583}{59535}, \quad \mu_8 = \frac{132809}{32531625}.$$

5 Twisted Poisson summation

Lemma 5.1 (Twisted Poisson identity). *For $m \in \mathbb{N}_{>0}$, $t \in \mathbb{R}$, and $z \in \mathbb{C}$,*

$$\sum_{k \in \mathbb{Z}} e^{kz} \phi\left(\frac{t-k}{m}\right) = m \sum_{n \in \mathbb{Z}} e^{(z-2\pi in)t} M(-mz + 2\pi imn). \quad (10)$$

Both sides are entire functions of z .

Proof. For fixed t , the left side is finite. Set $f(x) = e^{zx} \phi((t-x)/m)$. At integer frequency n ,

$$\begin{aligned} \widehat{f}(n) &= \int_{\mathbb{R}} e^{zx} \phi\left(\frac{t-x}{m}\right) e^{-2\pi inx} dx \\ &= m e^{(z-2\pi in)t} \int_{\mathbb{R}} \phi(u) e^{(-mz+2\pi imn)u} du \\ &= m e^{(z-2\pi in)t} M(-mz + 2\pi imn). \end{aligned}$$

Poisson summation applies because f is smooth and compactly supported; the rapid Fourier decay is locally uniform in z , giving an entire right side. $\square$

Let $\nu = \nu_2(m)$. At every nonzero frequency n , $M(2\pi imn)$ has zero order

$$1 + \nu_2(mn) \geq \nu + 1.$$

Consequently all nonzero aliases in (10) have zero Taylor jets through order ν at $z = 0$. This observation drives all exact polynomial identities below.

6 The up-Appell polynomials

The centered Appell inverse and its mean-value interpretation belong to the current repository corpus [3]; this section fixes the precise normalization needed for the interval synthesis and derives the coefficient forms used by the algorithm.

Definition 6.1 (Up-Appell sequence). For a positive integer m , define $A_r^{(m)}(u)$ by the germ at $z = 0$

$$\boxed{\sum_{r=0}^{\infty} A_r^{(m)}(u) \frac{z^r}{r!} = \frac{e^{uz}}{mM(mz)}}. \quad (11)$$

Proposition 6.2 (Algebraic properties). *For every $r \geq 0$,*

$$\frac{d}{du} A_r^{(m)}(u) = r A_{r-1}^{(m)}(u), \quad (12)$$

$$A_r^{(m)}(u+v) = \sum_{j=0}^r \binom{r}{j} A_j^{(m)}(u) v^{r-j}, \quad (13)$$

$$A_r^{(m)}(-u) = (-1)^r A_r^{(m)}(u), \quad (14)$$

$$A_r^{(m)}(mu) = m^{r-1} A_r^{(1)}(u). \quad (15)$$

The degree is r and the leading coefficient is $1/m$.

Proof. All statements follow directly from (11); parity uses the evenness of M . $\square$

6.1 Bell-polynomial formula

By Theorem 4.1,

$$\frac{e^{uz}}{mM(mz)} = \frac{1}{m} \exp \left(uz - \sum_{s \geq 1} \kappa_{2s} m^{2s} \frac{z^{2s}}{(2s)!} \right). \quad (16)$$

If Y_r is the complete exponential Bell polynomial, then

$$A_r^{(m)}(u) = \frac{1}{m} Y_r(u, -m^2 \kappa_2, 0, -m^4 \kappa_4, 0, \dots). \quad (17)$$

Only finitely many arguments matter for each r .

6.2 Triangular exact recurrence

Write

$$\frac{1}{M(mz)} = \sum_{n=0}^{\infty} \beta_n^{(m)} \frac{z^n}{n!}. \quad (18)$$

Multiplication by $M(mz) = \sum m^j \mu_j z^j / j!$ gives

$$\beta_0^{(m)} = 1, \quad \beta_n^{(m)} = - \sum_{j=1}^n \binom{n}{j} m^j \mu_j \beta_{n-j}^{(m)}. \quad (19)$$

Then

$$A_r^{(m)}(u) = \frac{1}{m} \sum_{j=0}^r \binom{r}{j} u^{r-j} \beta_j^{(m)}. \quad (20)$$

The full table through degree d costs $\mathcal{O}(d^2)$ rational operations.

6.3 The first universal polynomials

Put $P_r(u) = A_r^{(1)}(u)$. Then $A_r^{(m)}(u) = m^{r-1} P_r(u/m)$, and

$$\begin{aligned} P_0(u) &= 1, \\ P_1(u) &= u, \\ P_2(u) &= u^2 - \frac{1}{9}, \\ P_3(u) &= u^3 - \frac{1}{3}u, \\ P_4(u) &= u^4 - \frac{2}{3}u^2 + \frac{31}{675}, \\ P_5(u) &= u^5 - \frac{10}{9}u^3 + \frac{31}{135}u, \\ P_6(u) &= u^6 - \frac{5}{3}u^4 + \frac{31}{45}u^2 - \frac{2347}{59535}, \\ P_7(u) &= u^7 - \frac{7}{3}u^5 + \frac{217}{135}u^3 - \frac{2347}{8505}u, \\ P_8(u) &= u^8 - \frac{28}{9}u^6 + \frac{434}{135}u^4 - \frac{9388}{8505}u^2 + \frac{1904369}{32531625}. \end{aligned}$$

For example,

$$\begin{aligned} A_2^{(m)}(u) &= \frac{1}{m} \left(u^2 - \frac{m^2}{9} \right), \\ A_3^{(m)}(u) &= \frac{1}{m} \left(u^3 - \frac{m^2}{3} u \right), \\ A_4^{(m)}(u) &= \frac{1}{m} \left(u^4 - \frac{2}{3} m^2 u^2 + \frac{31}{675} m^4 \right). \end{aligned}$$

7 The global reproduction theorem

Theorem 7.1 (Exact monomial reproduction). *Let $m \in \mathbb{N}_{>0}$, $\nu = \nu_2(m)$, and $0 \leq r \leq \nu$. Then for every $t \in \mathbb{R}$,*

$$t^r = \sum_{k \in \mathbb{Z}} A_r^{(m)}(k) \phi\left(\frac{t-k}{m}\right). \quad (21)$$

The sum is locally finite.

Proof. In a neighborhood of $z = 0$, divide (10) by $mM(mz)$. The zero-frequency term is e^{tz} , because M is even. Every nonzero-frequency term vanishes to order at least $\nu + 1$. Therefore the Taylor coefficients through degree ν of

$$\sum_{k \in \mathbb{Z}} \frac{e^{kz}}{mM(mz)} \phi\left(\frac{t-k}{m}\right)$$

agree with those of e^{tz} . Apply (11) and compare the coefficient of $z^r/r!$. $\square$

Corollary 7.2 (Polynomial synthesis). *If $q \in \mathbb{R}[t]$ and $\deg q \leq \nu_2(m)$, define*

$$C_q^{(m)}(u) = \frac{1}{m} M(mD)^{-1} q(u). \quad (22)$$

Then

$$q(t) = \sum_{k \in \mathbb{Z}} C_q^{(m)}(k) \phi\left(\frac{t-k}{m}\right). \quad (23)$$

Proof. Linearity reduces to Theorem 7.1. The operator formula is equivalent to the generating function because $D^j e^{uz} = z^j e^{uz}$. $\square$

7.1 The inverse smoothing operator

On polynomial space,

$$M(mD)^{-1} = \exp\left(-\sum_{s \geq 1} \kappa_{2s} m^{2s} \frac{D^{2s}}{(2s)!}\right), \quad (24)$$

$$= \prod_{j=1}^{\infty} \frac{mD/2^j}{\sinh(mD/2^j)}. \quad (25)$$

The resulting power series is truncated at $D^{\deg q}$; its needed coefficients are convergent dyadic sums. In ordinary derivative notation,

$$M(\rho D)^{-1} p = p - \frac{\rho^2}{18} p'' + \frac{31\rho^4}{16200} p^{(4)} - \dots$$

Proposition 7.3 (Probabilistic characterization). *Let X have density ϕ . The polynomial $q^\sharp = M(mD)^{-1}q$ is the unique polynomial of degree at most $\deg q$ satisfying*

$$\mathbb{E}[q^\sharp(u + mX)] = q(u). \quad (26)$$

Equivalently,

$$m\mathbb{E}[A_r^{(m)}(u + mX)] = u^r.$$

Proof. For every polynomial f , $\mathbb{E}[f(u + mX)] = M(mD)f(u)$. The operator is triangular with unit diagonal in the monomial basis and is therefore invertible on each finite-dimensional polynomial space. $\square$

This explains the deconvolution terminology: the coefficient polynomial exactly undoes smoothing by a random up-distributed displacement.

8 Affine form and the finite-interval algorithm

Let $p \in \mathbb{R}[x]$, $x_0 \in \mathbb{R}$, $h > 0$, and $\rho = mh$. Apply [Theorem 7.2](#) to $q(t) = p(x_0 + ht)$.

Theorem 8.1 (Affine lattice formula). *Suppose $\deg p \leq \nu_2(m)$. Define*

$$p_\rho^\sharp = M(\rho D)^{-1}p = \exp\left(-\sum_{s \geq 1} \kappa_{2s} \rho^{2s} \frac{D^{2s}}{(2s)!}\right) p. \quad (27)$$

Then for every $x \in \mathbb{R}$,

$$p(x) = \frac{1}{m} \sum_{k \in \mathbb{Z}} p_\rho^\sharp(x_0 + kh) \phi\left(\frac{x - x_0 - kh}{\rho}\right). \quad (28)$$

Proof. In the variable $t = (x - x_0)/h$, one has $mD_t = mhD_x = \rho D_x$. Substitute the polynomial $q(t) = p(x_0 + ht)$ into (23). $\square$

Remark 8.2. The coefficient polynomial depends on the physical radius ρ , not separately on m and h . The ratio $m = \rho/h$ controls alias-zero multiplicity. Thus one may keep ρ fixed while taking the smallest admissible ratio $m = 2^{\deg p}$.

Theorem 8.3 (Finite exact polynomial plateau). *Let $I = [A, B]$, and retain only*

$$k_{\min} = \left\lceil \frac{A - x_0}{h} - m \right\rceil, \quad k_{\max} = \left\lfloor \frac{B - x_0}{h} + m \right\rfloor. \quad (29)$$

Then

$$P_I(x) = \frac{1}{m} \sum_{k=k_{\min}}^{k_{\max}} p_\rho^\sharp(x_0 + kh) \phi\left(\frac{x - x_0 - kh}{\rho}\right) \quad (30)$$

is compactly supported, belongs to $C^\infty(\mathbb{R})$, and satisfies

$$P_I(x) = p(x) \quad (x \in I).$$

Moreover,

$$\text{supp } P_I \subseteq [A - 2\rho, B + 2\rho], \quad (31)$$

and the listed atom count obeys

$$N \leq \left\lfloor \frac{B - A}{h} + 2m \right\rfloor + 1 = \left\lfloor m \left(\frac{B - A}{\rho} + 2 \right) \right\rfloor + 1. \quad (32)$$

Proof. An omitted atom has center farther than ρ from every point of I , so it vanishes on I . The finite sum therefore equals the global locally finite sum there. Retained centers lie in $[A - \rho, B + \rho]$; enlarging by the atom radius gives (31). Counting integers in (29) gives (32). $\square$

Corollary 8.4 (Jet matching). *For every $r \geq 0$,*

$$P_I^{(r)}(A) = p^{(r)}(A), \quad P_I^{(r)}(B) = p^{(r)}(B).$$

Derivatives of p above its degree are zero.

Proof. All derivatives agree in the interior of I , and both functions are smooth; take one-sided limits to the endpoints. $\square$

Thus the result is an exact polynomial plateau, not only a high-order approximation or a Hermite match at isolated points.

8.1 Constructive algorithm

Algorithm 8.5 (Exact common-scale representation). Input: $p(x) = \sum_{j=0}^d p_j x^j$, interval $[A, B]$, lattice origin x_0 , and either a step $h > 0$ or a radius $\rho > 0$.

1. Choose $m \in \mathbb{N}_{>0}$ with $\nu_2(m) \geq d$. The canonical minimum is $m = 2^d$. If ρ is prescribed, set $h = \rho/m$; if h is prescribed, set $\rho = mh$.
2. Compute $\kappa_2, \dots, \kappa_{2\lfloor d/2 \rfloor}$ from (8).
3. Apply the finite differential operator (27) to obtain $p_\rho^\sharp$. Equivalently, compute moments by (9), reciprocal coefficients by (19), and use (20).
4. Compute $k_{\min}, k_{\max}$ from (29).
5. For every integer k in this range, output

$$\frac{1}{m} p_\rho^\sharp(x_0 + kh) \phi\left(\frac{x - x_0 - kh}{\rho}\right).$$

The finite sum equals p identically on $[A, B]$.

Listing 1: Compressed pseudocode for the exact construction.

```

d := degree(p)
m := 2^d // or any m with v2(m) >= d
h := rho / m // when physical radius rho is chosen
for s from 1 to floor(d/2):
    kappa[2s] := 2^(2s-1)*Bernoulli(2s)/(s*(2^(2s)-1))
psharp := exp(-sum_s kappa[2s]*rho^(2s)*D^(2s)/(2s!)) applied to p
kmin := ceil((A-x0)/h - m)
kmax := floor((B-x0)/h + m)
return Sum(k=kmin..kmax,
           psharp(x0+k*h)/m * up((x-x0-k*h)/rho))

```

8.2 Compressed output and evaluation locality

The representation need not store N unrelated numbers. It is determined by the degree- d polynomial $p_\rho^\sharp$, the parameters (A, B, x_0, h, m) , and an integer range, requiring $\mathcal{O}(d)$ field elements. At a requested point x , only indices satisfying

$$\frac{x - x_0}{h} - m < k < \frac{x - x_0}{h} + m$$

contribute, so evaluation uses at most $2m + 1$ atoms independently of the length of I .

9 Sharp reproduction order and the first alias obstruction

The preceding construction proves reproduction through degree $\nu_2(m)$. The corpus's self-sampling theory already identifies the corresponding dyadic Strang–Fix mechanism and alias hierarchy [3]. We restate it in the present common-scale normalization because its sharp form supplies the feasibility lower bound and the first obstruction used later. For up, that order is sharp: a generator reproduces polynomials through degree d on a lattice only when its Fourier transform has zeros of order at least $d + 1$ at every nonzero dual-lattice point. By [Theorem 2.2](#), the smallest zero order among the samples $\widehat{\phi}(mn)$, $n \neq 0$, is $1 + \nu_2(m)$, attained exactly when n is odd.

Theorem 9.1 (First alias defect). *Let $m = 2^\nu q$, where q is odd, and put $L = \nu + 1$. If the same Appell coefficients are used one degree beyond the exact range, then*

$$\begin{aligned} E_{m,L}(t) &:= \sum_{k \in \mathbb{Z}} A_L^{(m)}(k) \phi\left(\frac{t-k}{m}\right) - t^L \\ &= \frac{(-1)^{L+1} L!}{(2\pi i)^L} \sum_{\substack{n \in \mathbb{Z} \\ n \text{ odd}}} \frac{\widehat{\phi}(qn/2)}{n^L} e^{-2\pi i n t}. \end{aligned} \quad (33)$$

The series is real-valued, smooth, one-periodic, and not identically zero.

Proof. Divide (10) by $mM(mz)$, as in the proof of [Theorem 7.1](#). The coefficient of $z^L/L!$ in the nonzero-frequency part is the defect. Frequencies with even n have zero order at least $L + 1$ and do not contribute. For odd n , exactly the factors indexed by $0, \dots, \nu$ vanish at mn . Expanding those simple zeros and retaining the remaining factors gives

$$\frac{1}{L!} \frac{d^L}{dz^L} \frac{M(-mz + 2\pi i m n)}{M(mz)} \Big|_{z=0} = \frac{(-1)^{L+1}}{(2\pi i n)^L} \widehat{\phi}(qn/2).$$

Multiplication by $L!e^{-2\pi i n t}$ yields (33). Pairing n with $-n$ proves reality. Rapid decay of $\widehat{\phi}$ permits termwise differentiation. Finally, the Fourier coefficient at $n = 1$ is nonzero because $q/2$ is not an integer zero of $\widehat{\phi}$; hence the defect is nonzero. $\square$

Corollary 9.2 (Exact stationary order). *For the common-scale lattice generated by $\phi((t-k)/m)$, the maximal exactly reproduced polynomial degree is*

$$\boxed{\nu_2(m)}.$$

Proof. [Theorem 7.1](#) gives the lower bound, while [Theorem 9.1](#) shows failure at the next degree. This is also the sharp Strang–Fix order associated with the minimum dual-lattice zero multiplicity. $\square$

Corollary 9.3 (Dyadic efficiency). *Among positive integer ratios m , exact reproduction of every polynomial of degree at most d requires and is achieved by*

$$m \equiv 0 \pmod{2^d}.$$

The smallest admissible ratio is therefore $m = 2^d$.

The result is arithmetically rigid: multiplying an admissible m by an odd integer increases overlap and atom count without increasing reproduction order. Multiplication by two increases the order by exactly one.

10 Arithmetic structure of the coefficients

Proposition 10.1 (Rational closure). *If $p, A, B, x_0, h \in \mathbb{Q}$ and $m \in \mathbb{N}_{>0}$, then every coefficient and center produced by Theorem 8.5 is rational. In particular, the algorithm is exact over $\mathbb{Q}$ and requires no evaluation of up to construct the representation.*

Proof. The Bernoulli cumulants (8), moments (9), reciprocal coefficients (19), and polynomial samples are rational. Floors and ceilings of rationals give exact integer bounds. $\square$

Proposition 10.2 (A simple dyadic-denominator bound). *Let $d = \deg p$, $\nu = \nu_2(m) \geq d$, and assume that the coefficients of p , together with x_0 and h , lie in $2^{-s}\mathbb{Z}[1/O]$, where O is odd. Then every output coefficient belongs to*

$$2^{-[s(d+1)+d+\nu]}\mathbb{Z}[1/O']$$

for some odd integer O' determined by d, O, m . Thus the power of two in all denominators is explicitly bounded; all further denominator growth is odd.

Proof. By Theorem 4.2, the cumulants have odd reduced denominators. In the truncated exponential (24), a term of total differential order at most d contains factorial denominators whose total 2-adic valuation is at most d . Evaluating a degree- d polynomial whose coefficients and argument have dyadic denominator at most 2^s contributes at most $s(d+1)$. Finally, the outer factor $1/m$ contributes ν . Adding these conservative bounds proves the assertion. A sharper valuation follows by applying Legendre's formula term by term in (17). $\square$

Remark 10.3. The proposition deliberately gives a robust rather than optimal exponent. For fixed degree, exact valuation minimization is a finite combinatorial problem in Bell-polynomial terms. The empirical denominators in the accompanying tables are substantially smaller.

11 Support, atom count, locality, and conditioning

11.1 Radius–density tradeoff

For the minimal choice $m = 2^d$, prescribing the physical radius ρ forces lattice spacing $h = \rho/2^d$. From (32),

$$N = \frac{2^d(B-A)}{\rho} + 2^{d+1} + \mathcal{O}(1). \quad (34)$$

Thus small support enlargement requires exponentially many atoms in the degree, whereas enlarging ρ reduces the number of interior centers but never the $2^{d+1} + \mathcal{O}(1)$ boundary-overlap term. The sum's support is always contained in $[A - 2\rho, B + 2\rho]$.

11.2 Coefficient and perturbation estimates

Define

$$K_\rho(p; A, B) = \sup_{y \in [A-\rho, B+\rho]} |p_\rho^\#(y)|.$$

Since every retained coefficient is $m^{-1}p_\rho^\#(x_k)$,

$$\sum_{k=k_{\min}}^{k_{\max}} |c_k| \leq \left(2 + \frac{B-A}{\rho} + \frac{1}{m}\right) K_\rho(p; A, B). \quad (35)$$

At each point only $2m+1$ terms can occur. Hence coefficient perturbations $|\delta c_k| \leq \varepsilon$ produce

$$\|\delta P_I\|_{L^\infty(I)} \leq (2m+1) \|\phi\|_\infty \varepsilon. \quad (36)$$

In the usual normalization $\phi(0) = 1$ and $0 \leq \phi \leq 1$, so the last factor may be omitted.

These bounds separate two notions of stability. The compressed polynomial data can be evaluated exactly or at high precision with only $\mathcal{O}(d)$ storage. A fully expanded coefficient vector may nevertheless have a large ℓ^1 -norm when $M(\rho D)^{-1}$ amplifies high-degree components. This is deconvolution conditioning, not a defect in the exact identity.

11.3 Operation counts

The cumulants through degree d require $\mathcal{O}(d)$ Bernoulli values. The reciprocal recurrence and Appell table cost $\mathcal{O}(d^2)$ exact field operations. Generating all N coefficients by Horner evaluation costs $\mathcal{O}(Nd)$, while a forward-difference table reduces this to $\mathcal{O}(d^2 + N)$ additions once the first $d + 1$ values are known. Querying the represented function at one point needs at most $2m + 1$ evaluations of up, independent of $B - A$.

12 A real-space antiderivative algorithm

The Fourier–Appell method is the cleanest direct construction. A second, complementary algorithm works entirely in real space and exposes a finite compatibility condition that is invisible in the global lattice formula. Write

$$U_{a,c}(x) = \phi\left(\frac{x-c}{a}\right), \quad a > 0.$$

Lemma 12.1 (Parent–child derivative identity). *For every $a > 0$ and $c \in \mathbb{R}$,*

$$DU_{2a,c} = \frac{1}{a}(U_{a,c-a} - U_{a,c+a}). \quad (37)$$

Proof. Apply (2) at $(x - c)/(2a)$ and use the chain rule. $\square$

Consider a finite child sum on an a -spaced lattice,

$$B(x) = \sum_{j=L}^R b_j U_{a,c+(2j+1)a}(x).$$

A parent sum $P(x) = a \sum_{j=L}^{R-1} p_j U_{2a,c+2(j+1)a}(x)$ satisfies $P' = B$ exactly when

$$b_L = p_L, \quad b_j = p_j - p_{j-1} \quad (L < j < R), \quad b_R = -p_{R-1}.$$

Lemma 12.2 (Finite discrete primitive). *Such a finite parent sum exists if and only if*

$$\sum_{j=L}^R b_j = 0. \quad (38)$$

When it exists, it is unique and

$$p_j = \sum_{r=L}^j b_r. \quad (39)$$

Proof. The displayed difference equations telescope. Their last boundary equation is equivalent to the zero-sum condition; the first equation then recursively determines every prefix sum. $\square$

12.1 Ghost atoms and constants of integration

A derivative representation known only on I need not satisfy (38). Place one or more child atoms with supports disjoint from I . They are identically zero on I but may carry the opposite total coefficient, making the enlarged finite sequence summable to zero. After applying Theorem 12.2, the parent sum is an exact antiderivative on I . Different ghost placements alter the additive constant without altering the derivative there. A single value condition fixes the desired constant.

For a block of prefix coefficients P_j , splitting the correcting ghost mass between the two ends adds a common scalar γ to all interior prefixes. Two useful canonical choices are

$$\gamma_2 = -\frac{1}{N} \sum_{j=1}^N P_j, \quad \text{minimizing } \sum_j |P_j + \gamma|^2, \quad (40)$$

$$\gamma_\infty = -\frac{\max_j P_j + \min_j P_j}{2}, \quad \text{minimizing } \max_j |P_j + \gamma|. \quad (41)$$

Thus ghost atoms provide not only existence but a small finite-dimensional stability optimization.

Algorithm 12.3 (Recursive real-space construction). To represent a polynomial p of degree d on I :

1. start from a finite partition-of-unity representation of the constant $p^{(d)}$ on a slightly enlarged interval;
2. add invisible ghost atoms so the finite coefficient sequence has sum zero;
3. take discrete prefix sums by (39) and replace pairs of child atoms by parents using (37);
4. select the ghost split to impose the required constant of integration;
5. enlarge the working interval and repeat d times.

The final finite C^∞ sum agrees with p on I .

This algorithm allows scales to change from one stage to the next. It is less compact than Theorem 8.5, but it is local, sparse, and naturally adapted to boundary constraints or variational coefficient selection.

13 The Thue–Morse derivative stencil

Iterating Theorem 12.1 produces a closed form whose signs are the signed Thue–Morse sequence.

Theorem 13.1 (Exact dyadic derivative stencil). *For every integer $d \geq 0$, $h > 0$, and $c \in \mathbb{R}$,*

$$\boxed{D^d U_{2^d h, c}(x) = h^{-d} 2^{-\binom{d}{2}} \sum_{k=0}^{2^d-1} (-1)^{\text{wt}_2(k)} U_{h, c+(2k-(2^d-1))h}(x).} \quad (42)$$

Proof. The case $d = 0$ is immediate. Differentiating once applies (37) to every current parent, doubles the number of leaves, and appends a binary digit whose value determines the new sign. At stage j , the chain-rule factor is $1/(2^j h)$. Their product is $h^{-d} 2^{-0-1-\dots-(d-1)} = h^{-d} 2^{-\binom{d}{2}}$. The final leaf center is the signed binary displacement shown in (42). $\square$

The cancellation of low moments follows from the Prouhet product

$$\sum_{k=0}^{2^d-1} (-1)^{\text{wt}_2(k)} y^k = \prod_{j=0}^{d-1} (1 - y^{2^j}), \quad (43)$$

which has a zero of order d at $y = 1$. Hence

$$\sum_{k=0}^{2^d-1} (-1)^{\text{wt}_2(k)} k^r = 0 \quad (0 \leq r < d).$$

The same algebra has a Gaussian-binomial deformation:

$$\prod_{j=0}^{d-1} (1 - yq^j) = \sum_{r=0}^d (-1)^r q^{\binom{r}{2}} \begin{bmatrix} d \\ r \end{bmatrix}_q y^r. \quad (44)$$

This suggests geometrically scaled atomic stencils, discussed among the conjectures below.

14 Finite sinc products and piecewise-polynomial generators

For $N \geq 1$, let

$$\widehat{\phi}_N(\xi) = \prod_{n=0}^{N-1} \text{sinc}\left(\frac{\pi\xi}{2^n}\right). \quad (45)$$

Its inverse Fourier transform ϕ_N is the density of a finite sum of centered uniform random variables. It is a compactly supported piecewise polynomial, with

$$\text{supp } \phi_N = [-(1 - 2^{-N}), 1 - 2^{-N}].$$

Indeed, the factor indexed by n is the transform of the uniform density on $[-2^{-n-1}, 2^{-n-1}]$, and the support radii add.

Theorem 14.1 (Reproduction order for a finite product). *Under (45), the integer-lattice system generated by $\phi_N((t - k)/m)$ has exact polynomial reproduction degree*

$$\min(N - 1, \nu_2(m)). \quad (46)$$

Proof. At a nonzero integer mn , the number of vanishing factors in (45) is $\min(N, 1 + \nu_2(mn))$. The least multiplicity over $n \neq 0$ is $\min(N, 1 + \nu_2(m))$. Subtracting one gives the sharp Strang–Fix degree. $\square$

For the same normalization, the moment product is $M_N(z) = \prod_{j=1}^N \sinh(z/2^j)/(z/2^j)$, and its cumulants satisfy the particularly simple exact truncation law

$$\kappa_{2r}^{(N)} = (1 - 2^{-2rN}) \kappa_{2r}. \quad (47)$$

Let $A_{r,N}^{(m)}$ be the corresponding Appell polynomial. For fixed r, m , expansion of the missing cumulant tail gives

$$A_{r,N}^{(m)}(u) = A_r^{(m)}(u) + \binom{r}{2} \kappa_2 m^2 2^{-2N} A_{r-2}^{(m)}(u) + \mathcal{O}_{r,m}(2^{-4N}). \quad (48)$$

Thus the finite piecewise-polynomial systems converge coefficientwise to the up-system at an explicitly dyadic rate, while retaining a finite exact reproduction range determined jointly by N and $\nu_2(m)$.

15 A worked exact degree-four example

Take

$$p(x) = \frac{7}{5} - 3x + 2x^2 - \frac{1}{2}x^3 + \frac{3}{7}x^4, \quad I = \left[-\frac{1}{2}, \frac{3}{2}\right]. \quad (49)$$

Choose the canonical ratio $m = 2^4 = 16$, physical atom radius $\rho = 2$, lattice step $h = \rho/m = 1/8$, and $x_0 = 0$. The finite deconvolution is

$$p_2^\#(y) = \frac{1301}{1575} - \frac{7}{3}y + \frac{6}{7}y^2 - \frac{1}{2}y^3 + \frac{3}{7}y^4. \quad (50)$$

Indeed, only the p'' and $p^{(4)}$ corrections in $M(2D)^{-1}$ survive.

The support test (29) gives

$$k_{\min} = -20, \quad k_{\max} = 28,$$

and hence 49 atoms:

$$p(x) = \sum_{k=-20}^{28} \frac{1}{16} p_2^\# \left(\frac{k}{8} \right) \phi \left(\frac{x - k/8}{2} \right), \quad -\frac{1}{2} \leq x \leq \frac{3}{2}. \quad (51)$$

The first six coefficients are

k	center $k/8$	coefficient
-20	$-5/2$	$460783/201600$
-19	$-19/8$	$203448071/103219200$
-18	$-9/4$	$10924481/6451200$
-17	$-17/8$	$149614271/103219200$
-16	-2	$31151/25200$
-15	$-15/8$	$108427271/103219200$

The complete list is in `example_coefficients.csv`.

The accompanying program evaluates both sides at 129 dyadic points of I using exact fractions. Fraction arithmetic for the coefficient identity. The maximum residual is the rational number 0. It also constructs generic examples of degrees 0, 1, ..., 6, with canonical ratios 1, 2, ..., 64; every one of the seven exact checks returns zero. Numerical values of up are used only to draw Figure 1, never to certify the algebra.

16 Nonuniqueness, canonicity, and null representations

Finite interval representations are highly nonunique.

- Any atom whose support misses I may be added with arbitrary coefficient.
- Any finite up-sum vanishing on I may be added.
- The parameters m, h, x_0, ρ may be changed, subject to $\rho = mh$ and $\nu_2(m) \geq d$.
- The real-space algorithm permits varying scales and different ghost placements.

The formula of Theorem 8.5 is nevertheless canonical after fixing (m, h, x_0) : the coefficient polynomial $m^{-1}M(\rho D)^{-1}p$ is uniquely determined. On the global stationary space, uniqueness follows from the generating-function calculation: the coefficient polynomial of degree at most d is the unique one whose up-average returns p , by Theorem 7.3. The finite interval truncation then keeps exactly the atoms that can affect I .

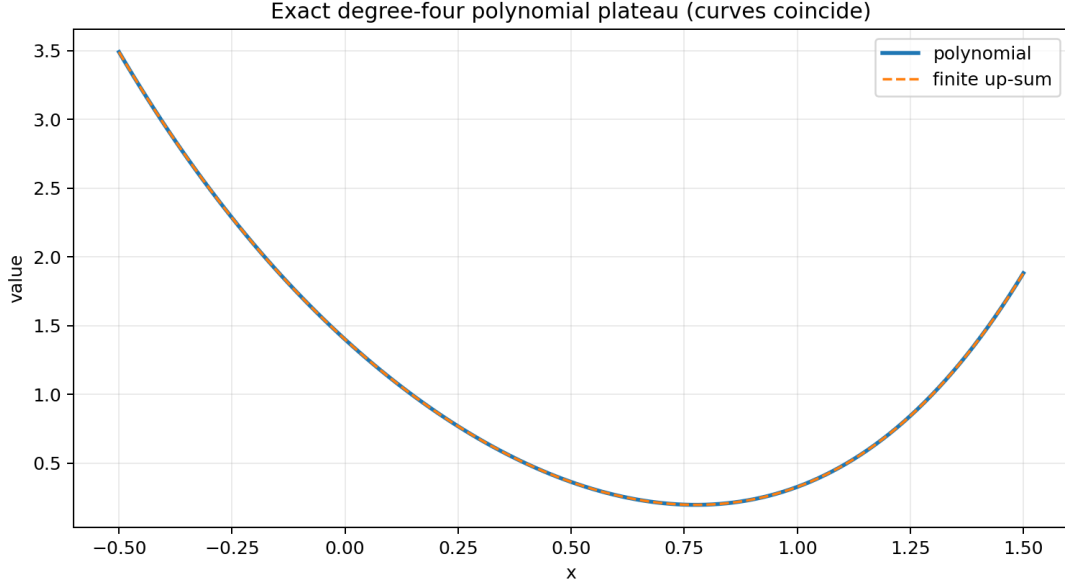


Figure 1: The target polynomial and its finite 49-atom up sum. The displayed numerical curves overlap on I ; the identity itself is exact and follows from [Theorem 8.3](#). Outside I , the compactly supported sum is free to leave the polynomial.

This distinction is useful computationally. One may regard the canonical formula as a particular solution and the finite sums vanishing on I as a nullspace. Additional objectives—minimum coefficient norm, prescribed symmetry, sparse scales, endpoint constraints, or integer coefficients after normalization—can then be posed as optimization in that nullspace without disturbing exactness.

17 A generator-independent theorem

The method is not peculiar to up; what is peculiar is the exact dyadic order law.

Theorem 17.1 (Compact-generator Appell reproduction). *Let $\psi \in C_c^\infty(\mathbb{R})$ satisfy $\int \psi = 1$, and put*

$$M_\psi(z) = \int_{\mathbb{R}} e^{zx} \psi(x) \, dx.$$

Fix $m \in \mathbb{N}_{>0}$. Suppose that, for every nonzero $n \in \mathbb{Z}$, $\widehat{\psi}$ has a zero of order at least $d + 1$ at mn . Define polynomials $A_{r,\psi}^{(m)}$ by

$$\sum_{r \geq 0} A_{r,\psi}^{(m)}(u) \frac{z^r}{r!} = \frac{e^{uz}}{m M_\psi(-mz)}. \quad (52)$$

Then, for $0 \leq r \leq d$,

$$t^r = \sum_{k \in \mathbb{Z}} A_{r,\psi}^{(m)}(k) \psi\left(\frac{t-k}{m}\right). \quad (53)$$

Consequently every degree- d polynomial has an exact finite representation on every compact interval after support truncation.

Proof. Repeat [Theorem 5.1](#) with ψ . Dividing by $m M_\psi(-mz)$ makes the zero-frequency term e^{tz} . The assumed zero multiplicities kill every nonzero-frequency Taylor coefficient through degree d . Coefficient comparison proves (53); compact support makes interval restriction finite. $\square$

For even ψ , $M_\psi(-z) = M_\psi(z)$, recovering the formulas used throughout this report. The theorem also clarifies the exact role of the infinite sinc product: its nonuniform integer zero multiplicities make the order depend on the binary valuation of the oversampling ratio.

18 Tensor-product multivariate extension

Let

$$\Phi(x_1, \dots, x_s) = \prod_{j=1}^s \phi(x_j).$$

Suppose a polynomial p has coordinate degree at most d_j in x_j . Choose m_j with $\nu_2(m_j) \geq d_j$, steps $h_j > 0$, radii $\rho_j = m_j h_j$, and origins x_{0j} . Define

$$p^\# = \prod_{j=1}^s M(\rho_j \partial_{x_j})^{-1} p. \quad (54)$$

The commuting differential operators terminate on p . Tensoring [Theorem 8.1](#) gives

$$p(x) = \frac{1}{\prod_j m_j} \sum_{\mathbf{k} \in \mathbb{Z}^s} p^\#(\mathbf{x}_0 + \mathbf{k} \odot \mathbf{h}) \prod_{j=1}^s \phi\left(\frac{x_j - x_{0j} - k_j h_j}{\rho_j}\right). \quad (55)$$

On a compact box only finitely many multi-indices contribute. The number of active atoms at one query point is at most $\prod_j (2m_j + 1)$. This gives exact smooth polynomial patches on boxes and, by affine coordinate changes, on parallelepipeds.

The tensor product is immediate but not necessarily optimal. A genuinely multivariate atomic generator with a Fourier zero set adapted to total degree could avoid the product-of-orders cost. This is a natural direction for higher-dimensional Rvachev finite elements.

19 Further structural consequences

19.1 Exact smooth localization of polynomial germs

For every compact interval I , polynomial p , and $\rho > 0$, there is a compactly supported C^∞ function in a finite span of equal-scale up-atoms that equals p on I and is supported in $[A - 2\rho, B + 2\rho]$. Unlike multiplication by a generic bump function, the construction remains inside a prescribed atomic family and gives all coefficients exactly.

19.2 Why the mechanism is polynomial-specific

The nonzero aliases in (10) have high-order zeros at $z = 0$, not identically zero germs. Consequently a finite Taylor jet is reproduced exactly, while an exponential $e^{\lambda t}$ generally retains nonzero aliases at $z = \lambda$. The report therefore does not assert exact exponential reproduction, nor impossibility for arbitrary varying-scale finite sums; it isolates the precise stationary obstruction.

19.3 Piecewise-polynomial targets

A finite sum of up-atoms is C^∞ . It cannot equal a target with a genuine derivative jump at an interior knot. Separate polynomial pieces can, however, be represented exactly on closed subintervals separated by transition gaps, or combined with explicitly chosen smooth transition zones. If all derivatives match at every knot, a finite piecewise polynomial on a connected interval is already one polynomial, and the main theorem applies directly.

19.4 Moment and derivative duality

The Appell construction and the Thue–Morse stencil are two sides of one operator identity. The Appell coefficients invert the smoothing operator $M(\rho D)$, whose logarithm is an even cumulant series. The real-space stencil factors derivatives into dyadic parent–child differences. In Fourier space these become, respectively, division by the sinc product near the origin and multiplication by its zero factors at dyadic frequencies. Bell polynomials organize the former; Prouhet–Thue–Morse cancellation organizes the latter.

20 Conjectures and frontier questions

The following statements are plausible consequences or extensions of the proved theory, but they are not established here.

Conjecture 1 (interval-specific dyadic lower bound). Fix $d \geq 1$. There is a length threshold L_d such that, if a degree- d polynomial is represented on an interval of length at least $L_d h$ by a finite sum of one-scale atoms $\phi((x - kh - x_0)/(mh))$ with arbitrary coefficients and with no atoms identically zero on the interval, then $\nu_2(m) \geq d$. Thus sufficiently long exact plateaux cannot beat the stationary binary order by boundary engineering.

Conjecture 2 (exponential atom complexity). For intervals whose length is comparable to the atom radius and under a uniform bound on support enlargement, every one-scale exact representation of all degree- d polynomials requires $\Omega(2^d)$ atoms in the worst case. The canonical construction, with $2^{d+1} + \mathcal{O}(1)$ boundary atoms, would then be order-optimal.

Conjecture 3 (stable-radius principle). For each fixed polynomial and interval, the coefficient norm of the canonical representation has a finite minimizing radius $\rho_* > 0$. After affine normalization of the interval, ρ_* is controlled by the nearest complex critical scale of the inverse symbol $1/M(\rho z)$, and varies continuously with the polynomial away from degree drops.

Conjecture 4 (ghost/Appell variational limit). Apply the real-space antiderivative algorithm on successively larger symmetric working intervals and choose the ℓ^2 -optimal ghost split (40) at every level. After matching the final scale and lattice, the interior coefficient sequences converge to the canonical Appell samples of [Theorem 8.5](#).

Conjecture 5 (local extended-Chebyshev behavior). For generic ordered centers and a sufficiently short interval avoiding all support endpoints, a finite family of equal-scale up-translates forms an extended Chebyshev system. If true, local interpolation matrices would be strictly sign-regular and the null representations discussed in [Section 16](#) would arise only after the interval crosses a generator-dependent length threshold.

Conjecture 6 (sharp odd-denominator law). After removing the evident factor m^{r-1} , the reduced denominator of $A_r^{(m)}(u)$ at dyadic u has a 2-adic valuation determined solely by the dyadic denominator of u and the binary digit sums of the Bell partitions of r . All irregular prime growth is confined to odd primes dividing numbers $2^{2^j} - 1$ and Bernoulli numerators.

Conjecture 7 (endpoint-scale lower-Lambert asymptotics). Let $P_d(u) = A_d^{(1)}(u)$. In the boundary scaling $u = 1 + \delta_d$, with $\delta_d 2^d/d$ bounded away from both zero and infinity, the dominant positive-real saddle z_d obeys

$$d = \delta_d z_d + \log_2 z_d + \Theta(1).$$

The full asymptotic expansion of $P_d(1 + \delta_d)$ contains a nonconstant one-periodic function of the corresponding lower-Lambert phase, analogous to the periodic correction in the Fabius endpoint asymptotic. Fixed- u coefficients, by contrast, are governed by the nearest imaginary poles of $1/M$.

Conjecture 8 (q -geometric atomic reproduction). There is a nonstationary atomic family with geometrically varying scales whose exact derivative and reproduction stencils have coefficients $(-1)^r q^{\binom{r}{2}} \left[\begin{smallmatrix} d \\ r \end{smallmatrix} \right]_q$. At $q = 2$ it should interact with the Rvachev refinement equation, while the limit $q \rightarrow 1$ recovers ordinary finite differences and classical Appell reproduction.

21 Promising research directions

21.1 Lean formalization

A compact formalization path is:

1. prove the exact integer zero multiplicity of the sinc product;
2. formalize twisted Poisson summation for a smooth compactly supported function with an exponential twist;
3. define the finite-degree inverse $M(mD)^{-1}$ in $\mathbb{Q}[D]/(D^{d+1})$;
4. prove the Appell generating identity and coefficient comparison;
5. derive the affine finite-interval truncation and support bounds;
6. formalize the parent-child derivative identity, finite prefix-sum primitive, and Thue-Morse stencil.

Most analytic difficulty is concentrated in Poisson summation; the remaining steps are finite polynomial algebra and exact support reasoning.

21.2 Certified symbolic software

The Python prototype should be developed into a small exact library that: accepts rational or algebraic input; emits a compressed coefficient polynomial or an expanded atom table; optimizes ρ under support and norm constraints; evaluates up at dyadic points with the repository's exact algorithm; and exports Lean-checkable certificates consisting of moment recurrences, polynomial identities, and support inequalities.

21.3 Minimal varying-scale representations

The common-scale theorem is explicit but probably not atom-minimal. The real-space construction suggests a mixed-integer problem over a dyadic tree: select parent atoms, ghost leaves, and scale transitions so that derivative constraints telescope while minimizing atom count or coefficient norm. Lower bounds could combine local linear independence, support geometry, and Fourier zero multiplicities.

21.4 Atomic finite elements and exact patch tests

Polynomial reproduction is the mathematical core of a finite-element patch test. Tensor products of the present plateaux give compactly supported C^∞ shape-function combinations that pass every prescribed polynomial patch test exactly. Useful next questions include sparse multivariate generators, boundary-adapted bases, exact quadrature, stiffness matrix arithmetic, and condition numbers relative to conventional high-order splines.

21.5 Asymptotics of the Appell table

The generating function $e^{uz}/M(z)$ offers several asymptotic regimes: nearest-pole expansions for fixed u , saddle-point behavior when u grows with degree, and the delicate endpoint scale $u = \pm 1 + \mathcal{O}(d2^{-d})$, where the exponential type of M nearly cancels the numerator. Mellin analysis of $\log M$, inverse lower-Lambert coordinates, and periodic fluctuations from the dyadic product should connect this coefficient problem directly to the repository's endpoint machinery.

22 Summary of proved results

For every polynomial p of degree d , every compact interval $I = [A, B]$, every $x_0 \in \mathbb{R}$, and every $\rho > 0$, choose $m = 2^d$ and $h = \rho/m$. Then

$$p(x) = \frac{1}{m} \sum_{k=\lceil (A-x_0)/h-m \rceil}^{\lfloor (B-x_0)/h+m \rfloor} [M(\rho D)^{-1}p](x_0 + kh) \operatorname{up}\left(\frac{x-x_0-kh}{\rho}\right) \quad (x \in [A, B]).$$

The sum is finite, uses one common scale, has rational coefficients for rational data, matches every endpoint jet, and is supported in $[A - 2\rho, B + 2\rho]$. Its atom count is at most

$$\left\lfloor m \left(\frac{B-A}{\rho} + 2 \right) \right\rfloor + 1.$$

For an arbitrary positive integer m , the exact stationary reproduction degree is precisely $\nu_2(m)$; the next degree has the explicit nonzero alias defect (33). The inverse symbol is governed by the Bernoulli cumulants (8), while repeated real-space differentiation produces the Thue–Morse stencil (42).

A Exact dyadic values used for independent checks

The numerical plot evaluates up through the bounded Fabius function F , using $\operatorname{up}(x) = F(1 - |x|)$ for $|x| \leq 1$. One exact dyadic formula compatible with the repository normalization is

$$F\left(\frac{a}{2^n}\right) = \frac{2^{-\binom{n+1}{2}}}{n!} \sum_{j=0}^{a-1} (-1)^{\operatorname{wt}_2(j)} \sum_{r=0}^{\lfloor n/2 \rfloor} \binom{n}{2r} \mu_{2r}(2a - 2j - 1)^{n-2r}, \quad (56)$$

for $0 \leq a \leq 2^n$, with the endpoint conventions $F(0) = 0$, $F(1) = 1$. Reduction to lowest terms is performed after the integer and moment sums are complete. This formula supplies exact reference values; the proof of polynomial reproduction itself does not depend on pointwise evaluation of F or up .

B Archive manifest and reproducibility

The ZIP archive contains:

File	Purpose
Exact_Polynomial_Plateaux_from_Rvachev_Up.tex	Complete LaTeX source of this report.
Exact_Polynomial_Plateaux_from_Rvachev_Up.pdf	Rendered report.
rvachev_polynomial_reproduction.py	Commented exact-arithmetic construction, verification, CSV generation, and plotting code.
example_coefficients.csv	All 49 rational coefficients of (51).
exact_grid_check.csv	Exact values and residuals at 129 dyadic test points.
appell_coefficients_m16.csv	Coefficient table for $A_r^{(16)}$, $0 \leq r \leq 8$.
generic_degree_checks.csv	Independent exact checks for degrees 0 through 6.
verification_summary.txt	Human-readable zero-residual summary.
representation_plot.png	Figure used in Figure 1.
README.md	Reproduction instructions and normalization notes.
SHA256SUMS	Integrity hashes for the packaged files.

Run

```
python rvachev_polynomial_reproduction.py
latexmk -pdf -interaction=nonstopmode \
    -halt-on-error Exact_Polynomial_Plateaux_from_Rvachev_Up.tex
```

from the unpacked directory. The script uses only the Python standard library for exact algebra; Matplotlib is used for the optional figure.

C Notation crosswalk

Symbol	Meaning
$\phi = \text{up}$	Even Rvachev up-function supported on $[-1, 1]$.
F	Bounded Fabius distribution function on $[0, 1]$, extended by constants outside.
$M(z)$	Moment-generating function $\int e^{zx} \phi(x) dx$.
κ_{2r}	Even cumulants of the probability density ϕ .
$A_r^{(m)}$	Up-Appell polynomial defined by (11).
h	Lattice spacing between atom centers.
m	Integer radius-to-spacing ratio.
$\rho = mh$	Physical radius/scale of each atom.
$p_\rho^\sharp$	Exact deconvolution $M(\rho D)^{-1}p$.
$\text{wt}_2(k)$	Number of one-bits in the binary expansion of k .
$\nu_2(k)$	Exponent of two dividing a nonzero integer k .

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